Physical and Behavioral Sequelae of Low Birthweight in Spiny Mice (*Acomys cahirinus*)

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Physical and behavioral sequelae of low birthweight (LBW) were investigated in spiny mice *(Acomys cahirinus)*. When raised by their biological mother, pups whose Day-2 weights were 1 SD or more below the population mean remained smaller through the first 30 days postpartum than pups born within normal (N) weight ranges. LBW pups fostered onto mothers who gave birth to N pups gained weight more rapidly than LBW pups remaining with their own mother. Similarly, N pups fostered onto mothers of LBW pups displayed reduced weight gain relative to N siblings raised by their biological mother. It appears that rate of weight gain is influenced by an interaction between birthweight and maternal rearing environment. LBW pups also were less responsive than N pups to maternal chemical cues, indicating that low birthweight is correlated with deficits in adaptive behavior.

For both mammalian neonates and avian hatchlings, low body weight may be predictive of subsequent developmental anomalies or reduced likelihood of survival. In oystercatchers, for example, subordinate chicks tend to weigh less than their dominant siblings and are more vulnerable to predation and starvation during periods of limited food supply (Safriel, 1981). Among avian species that begin incubation before the entire clutch of eggs is produced, asynchronous hatching results in a wide range in the size of chicks within a given brood (Mock, 1985). Older, larger chicks often receive a disproportionate share of food from their parents and may even attack and kill their smaller siblings (Ingram, 1959; Mock, 1985; O'Connor, 1978). Studies of rabbits and dogs reveal that physically immature neonates suffer relatively high mortality rates, apparently because of "limited ability to utilize the mother's milk" (Arshavsky & McDonald, 1968). Similarly, cannibalism of low-birthweight laboratory mice has been reported to be more common than for pups within normal weight ranges (Gandelman & Simon, 1977). In an extensive study of factors influencing development of pigtail monkeys, low-birthweight and premature animals displayed delays in reflex development, growth rate, and achieving major developmental milestones (Sackett & Holm, 1980). Furthermore, anomalous phys-

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iological measures (namely, respiration, cardiac rate, and diurnal temperature cycles) were evident throughout the first month of life in low-birthweight animals, and deviant behavior was observed at 3–4 months of age. Low birthweight in humans is likewise correlated with the development of a variety of neurological and behavioral abnormalities (Niswander & Gordon, 1972) ranging from learning disabilities (Hunt, Tooley, & Harvin, 1982) and delays in perceptual, memory, and motor abilities (Siegel, 1982) to hyperkinesis and autism (Caputo & Mandell, 1970).

To a great extent, our current understanding of behavioral and physiological development in low-birthweight rodents is based upon experiments in which birthweight is influenced by perinatal manipulations of the mother or offspring. Accordingly, rat pups that were surgically delivered 18 hr prematurely weighed less than full-term pups and were delayed in the development of negative geotaxis (Grota & Ambruso, 1973). Female rats and mice that are malnourished during pregnancy or after delivery consistently produce offspring characterized by stunted physical growth and delays in behavioral development (Frankova, 1973; Levitsky & Barnes, 1972; Smart, 1983). Since low birthweight is a correlate of the primary independent variables being assessed in such studies, however, one usually cannot draw conclusions concerning the influence of birthweight per se on subsequent development.

Behavioral and physical sequelae of low birthweight were investigated further in the following series of studies of laboratory-reared spiny mice *(Acomys cahirinus)*. Rather than assessing the development of animals whose low birthweight was the result of specific experimental manipulations, *A. cahirinus* subjects were drawn from pups whose *naturally occurring* weights were below the population mean. *A. cahirinus* is unique among murid rodents in giving birth to small litters of precocial offspring after a gestational period of approximately 38 days. Concomitant with the relatively large size of the precocial neonates, there is greater variability in their body weights than is found in small altricial rodents. The following studies are concerned specifically with weight gain and its susceptibility to postnatal maternal influences, as well as the development of responsiveness to social odors, in low-birthweight pups as compared with pups whose weights fall within a statistically normal range.

BIRTH WEIGHTS AND WEIGHT GAIN THROUGH DAY 30 POSTPARTUM

Subjects

Seventy-five litters, each containing from two to five pups born in the laboratory breeding colony of *Acomys cahirinus,* were randomly selected for inclusion in this study. Each newborn litter was housed in a separate individual breeding cage along with both parents throughout the 30-day test period. Only one litter of pups was included for any given mated pair.

Procedure

Two pups were selected at random for each litter on the day after birth (Day 2) and individually weighed to the nearest 0.1 g. Immediately after this initial weight was recorded, the two littermates were ear-punched to allow for individual identification and replaced into their home cage. Pups were weighed for the first time on Day 2 rather than on the day of birth to avoid disrupting the recently parturient female while she was cleaning her neonates or before she had delivered the entire litter. In this manner, the same two pups from each litter were weighed at weekly intervals over the first 30 days postpartum (i.e., on Days 2, 9, 16, 23, and 30). Aside from these repeated weighings, pups were not otherwise disturbed during the course of this study.

Results

Variability in Day-2 Body Weights. In three litters, only one pup survived until the second day postpartum. Therefore, Day-2 weights were obtained for a total of 147 pups. The mean body weight for these pups was 6.4 g (SD = 1.0 g) on Day 2.

Weight Gain through Day 30. Pups were assigned to three categories as a function of their Day-2 body weight:

1. Pups whose body weight ≥ 1 SD above the mean (≥ 7.4 g; n = 23).
2. Body weight 1 SD or more below the mean (≤ 5.4 g; n = 20).
3. Body weight within ± 1 SD of the mean (5.5–7.3 g; n = 104).

Weight gain by pups in these three categories is presented in graphic form in Figure 1, which shows mean weights on Day 2 and after each of the one-week intervals through Day 30. As is evident on inspection of Figure 1, *A. cahirinus* pups tended to maintain their relative rankings on body weight from days 2 through 30 postpartum. That is, pups whose body weights were 1 SD or more above the mean on Day 2 also tended to have relatively high weights at each of the weekly weighing sessions. Similarly, over this same time period, the low-birthweight pups (i.e., those whose Day-2 weights were at least 1 SD below the mean) remained lighter than the pups in the other two conditions.

POSTNATAL MATERNAL INFLUENCES ON WEIGHT GAIN BY SUCKLING PUPS

In the previous study, all pups, regardless of their initial body weights, were reared by their biological mothers. It is therefore not possible to ascertain whether the continued disparity in body weights across the three conditions through Day 30 was a function of intrinsic characteristics of the pups themselves, differential maternal influences among the three conditions, or an interaction of these two variables. Furthermore, since all pups in each litter were not systematically weighed (i.e., data were only obtained on two pups

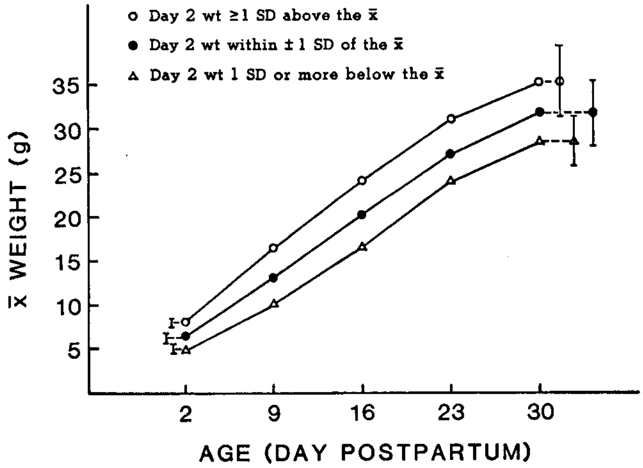

Fig. 1. Mean weights (at one-week intervals) on Day 2–30 postpartum for pups in the three Day-2 weight conditions (vertical bars = ± 1 SD).

per litter), the results could reflect weight gain of pups from heterogeneous litters (those containing pups with a high degree of variability in Day-2 body weights) as well as litters made up entirely of pups within a single weight category. The next experiment in this series was therefore designed to elucidate the extent to which weight gain in low-birth weight pups and those born within (or above) normal weight ranges is susceptible to influences from the postnatal maternal environment. In each instance, only litters composed of pups of relatively homogeneous birth weights were included in this study.

Methods

Body-Weight Categories. Pups born in the *A. cahirinus* breeding colony were systematically weighed on Day 2. A total of 26 litters meeting the following weight criteria were included in this experiment:

Low-birthweight litters (n = 13): Litters defined as low birthweight (LBW) originally contained 2–5 pups, with the majority of neonates in a litter each weighing 5.4 g or less on Day 2. In addition, the mean Day-2 weights of pups comprising a low-birthweight litter $\leq$ 5.4 g.

Normal-weight litters (n = 13): Litters were designated as normal weight (N) if the majority of pups weighed at least 5.5 g each and the mean pup weight $\geq$ 5.5 g.

Following the initial weighings on Day 2, all litters (LBW as well as N) were culled to two pups each. These remaining two pups were ear punched for individual identification. In each instance, individual body weights of the two pups in an LBW litter were no more than 5.4 g (range = 3.2–5.4 g; $\overline{X}$ = 4.4), whereas the two pups remaining in each N litter weighed a minimum of 5.6 g (range = 5.6–8.1; $\overline{X}$ = 6.4).

Cross-Fostering Procedure. On Day 2, after the initial body weights were obtained, one of the two pups in an LBW litter was chosen at random and switched with an N pup (also randomly selected) of the same age. At that time, an LBW pup was placed into the home cage of an N litter, while a single pup from the latter litter was removed and placed into the home cage of the LBW pup. The two litters involved in each exchange of pups had been born on the same day. Following this cross-fostering manipulation, each LBW home cage contained one of the original LBW pups, one foster N pup, and both parents of the LBW pup. In a similar manner, each of the N family cages housed one N pup and its parents along with a single LBW foster pup. The four animals in each home cage were then housed together throughout the 30-day testing period. Thus, each of the 13 mothers who gave birth to LBW litters reared 2 pups; one of her own LBW pups and one foster N pup. Likewise, all 13 mothers who gave birth to N litters reared one of those pups along with one LBW foster pup. The two pups in each cage (one original and one foster) were then weighed at weekly intervals through Day 30 (i.e., on Days 9, 16, 23, and 30).

Results

Mean weights of pups on Days 2 through 30 are presented in Figure 2 for the four treatment groups (namely, *Foster* and *Own* pups raised by LBW mothers and *Foster* and *Own* pups raised by N mothers). A 4 (treatment group) x 4 (day of weighing; repeated measures) ANOVA compared pup body weights at weekly intervals beginning on Day 9. Day 2 weights were not included in the statistical analyses since pups were assigned to either the LBW or N conditions according to their weights at that age, and the primary question addressed in this study concerned weight gain as a function of rearing environment following the Day 2 fostering manipulations. Statistically reliable main effects were obtained for treatment condition (F = 16.00; df ⅓; $p < .0001$) and day of weighing

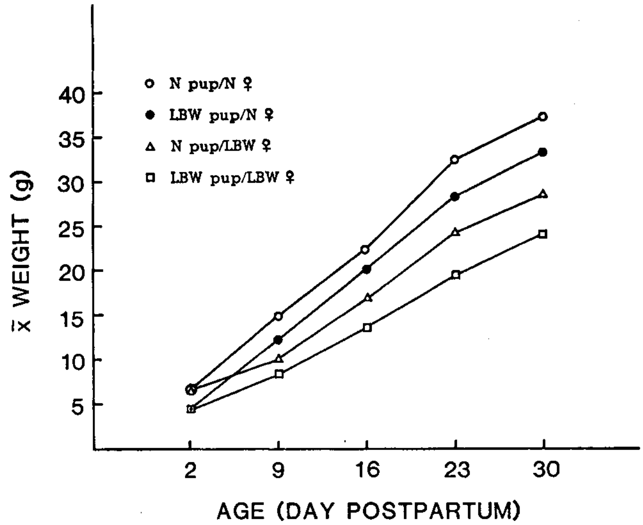

Fig. 2. Mean weights (at one-week intervals) of LBW and N pups reared by LBW versus N mothers on Day 2–30 postpartum.

(F = 877.08; df 1/3; $p < .0001$), as well as the day × treatment interaction (F = 5.73; df 9/114; $p < .0001$). Post-hoc analyses (Newman Keuls) further revealed that the mean weight (collapsed across Days 9–30) of N pups reared by their own mother was significantly *greater* ($p < .05$) than that of pups in the other three treatment conditions; similarly, LBW pups reared by their own mother weighed significantly *less* than pups in the other treatment conditions. LBW pups fostered onto N mothers did *not* differ reliably in mean weight from N pups reared by LBW mothers. Since the growth curves of both the LBW and N offspring differed according to whether they were raised by an N or LBW mother, it is evident that the maternal rearing environment has a significant influence on the rate of pups' postnatal weight gain.

RESPONSES OF LOW-BIRTHWEIGHT PUPS TO MATERNAL CHEMICAL CUES

Within the first 36 hr after birth, normally developing spiny mouse pups begin to orient and locomote preferentially toward chemical cues produced by lactating females (Porter & Doane, 1976; Porter & Ruttle, 1975). Such early attraction to a maternal pheromone presumably functions to keep the mobile precocial neonates from wandering away from their mother or home area and may therefore be critical for survival of preweanling pups. The final experiments in the present series assessed whether low-birthweight pups would display any noticeable deficit in this early form of adaptive behavior.

Day-2 Tests

Methods

Subjects: Forty-four *A. cahirinus* pups were each given a series of bedding-preference tests on Day-2 postpartum. For 22 of these pups (LBW), individual Day-2 body weights were 1 SD or more below the mean of 6.4 g established in the first study reported above ($\bar{X}$ = 4.4 g; range = 3.7–5.2). The remaining 22 pups (N) had Day-2 body weights within 1 SD of the previously established mean ($\bar{X}$ = 6.2 g; range = 5.6–7.2).

No more than three pups from any given litter were included in this experiment, and they all remained in their original home cage along with both parents and littermates until the beginning of the test session.

Apparatus: Testing was conducted in a plastic cage measuring 28 × 18 × 13 cm high with shallow layers of bedding material (sugar cane waste) placed onto the floor at opposite ends of the cage. The two layers of bedding completely covered the floor of the test cage except for a 4-cm-wide clean starting strip centered between them. Bedding material on one side of the starting strip had been taken from soiled areas of the subject pup's own home cage immediately prior to the test session. The floor on the opposite side of the starting strip was covered with clean bedding material.

Testing procedure: At the beginning of each test trial, a single pup was placed into the center of the uncovered starting area facing at a 90° angle to the two areas of bedding material and observed until it moved entirely onto one of the areas of bedding or until 120 sec had elapsed without a choice response being made. At that time, the animal was removed from the test cage and placed into a holding chamber while the test cage was rotated 90°. As soon as the test cage was in its new position, the subject pup was replaced into the starting area for a second choice test. Seven consecutive bedding-choice tests were conducted in this manner for each animal on Day 2 postpartum.

Results

Mean responses to soiled versus clean bedding by both the N and LBW pups are presented in Table 1. A single preference score was calculated for the seven consecutive bedding-choice tests for each pup by subtracting the number of times that it moved onto the clean bedding material from the number of times that it chose the soiled bedding from its own home cage. Wilcoxon's matched-pairs signed-ranks tests of these preference scores indicate that both the LBW and N pups selected the soiled bedding significantly more often than the clean bedding material (see Table 1). Additional comparisons *between* the two conditions revealed that the total number of choice responses (i.e., excluding trials when the animal did not leave the starting strip) was significantly greater for the N pups ($\bar{X}$ = 6.9) than for the LBW pups ($\bar{X}$ = 6.3) ($Z = 2.27$, $p < .025$, Mann-Whitney U test). Also, the LBW pups chose the *clean* bedding material significantly more often than did the N pups ($Z = 2.10$, $p < .05$, Mann Whitney U test).

Day-9 Tests

As evident from the results of the previous experiment, both LBW and N pups are attracted to soiled bedding from their own home cage on Day 2 postpartum. In comparison to N pups, however, LBW pups make more inappropriate responses (i.e., LBW pups

TABLE 1. Summary of Results of Choice Tests for Home-cage Bedding versus Clean Bedding for both Low-birthweight (LBW) and Normal-weight (N) Pups Tested on Day 2 Postpartum.

	Pup Condition	
	LBW	N
$\bar{X}$ responses to homecage (soiled) bedding	5.05	6.42
$\bar{X}$ responses to clean bedding	1.23	0.50
$\bar{X}$ preference score	3.82	5.91
p (Wilcoxon's test)	<.001	<.001
	($t = 3$; n-ties = 21)	($t = 0$; $n = 22$)

chose clean bedding more frequently than did the N pups). The final experiment in this series was conducted to determine whether LBW pups display any noticeable deficit in responsiveness to maternal pheromone at an older age, namely, Day 9 postpartum. Accordingly, in contrast with the stimulus bedding used in the Day-2 tests, pups in the present experiment were given a choice between bedding soiled by a lactating female versus bedding soiled by a nonlactating (nulliparous) adult female. Previous research in our laboratory has determined that normally developing *A. cahirinus* pups continue to respond to chemical cues produced by lactating females until the approach of weaning (Porter, Doane, & Cavallaro, 1978).

Methods

Subjects: Sixteen low-birthweight (LBW) and sixteen normal-birthweight (N) pups were identified on the second day postpartum for inclusion in this study. The mean Day-2 body weight of the LBW pups was 4.2 g (range = 3.5–5.2), and that of the N pups was 6.1 g (range = 5.5–7.5). All 32 pups remained in their respective home cages along with their littermates and both parents until testing on Day 9 postpartum. At that time, the 16 LBW pups had a mean body weight of 9.8 g (range = 6.5–12.2) compared with a mean of 12.8 g (range = 7.0–17.2) for the 16 N pups.

Apparatus: Because of the substantial increase in size and mobility of *A. cahirinus* pups over the first few days after birth, the apparatus used for the Day-9 tests differed from that employed in the Day-2 tests reported above. All Day-9 test trials were conducted in a glass terrarium measuring 30 × 58 × 30 cm high. Two galvanized metal inserts partitioned the terrarium into two 30 × 23 cm "stimulus choice" areas separated by a 30 × 12 cm-wide clean starting area. An 8 × 8 cm-high opening centered at the bottom of each partition allowed complete access into all areas of the testing terrarium. One of the stimulus choice areas was covered with a shallow layer of bedding material that had been soiled by a *lactating* female (who was not the mother of any of the test pups). Each of the stimulus lactating females had been housed with her own 7- to 10-day-old suckling litter until being removed from her home cage and housed in an isolation cage on clean bedding material for approximately 24 hr prior to the test session. Immediately before the beginning of the test session, soiled bedding was removed from this isolation cage and placed on the floor of one of the choice areas in the testing terrarium. The opposite stimulus-choice area was covered with bedding that had been soiled for approximately 24 hr by a *nonlactating* (nulliparous) adult female.

Testing procedure: Each pup was given seven consecutive choice tests for its responses to bedding soiled by a lactating female versus a nonlactating female. A test trial commenced when a pup was placed into the starting area so that its body faced parallel to the two partitions. The animal was then observed until it moved entirely (i.e., all four paws) into one of the choice areas, or until 120 sec had elapsed without a choice being made. No more than two animals were tested with the same stimulus bedding material, and the testing apparatus was dismantled, washed, and rinsed before introduction of new stimulus bedding and testing of subsequent animals. Further details of the testing procedure are otherwise identical to those described above for the Day-2 tests.

Results

A preference score was calculated for each pup by subtracting the number of times it chose the bedding soiled by a nonlactating female from the number of responses to the lactating female bedding. Within each condition (LBW and N) the frequency of choices of bedding soiled by lactating versus nonlactating females were then compared with Wilcoxon's matched-pairs signed-ranks test. The results of these analyses (summarized in Table 2) indicate that the N pups significantly preferred bedding material soiled by a lactating female. LBW pups, in contrast, displayed no significant difference in the frequencies of their responses to the two odor stimuli. Comparisons *between* the two weight conditions revealed that the N pups made significantly more choice responses (total of lactating female choices plus nonlactating female choices) than did the LBW pups (means = 6.1 and 4.7, respectively; $Z = 2.33$, $p < .025$, Mann-Whitney U test).

Discussion

As seen in the results of the normative study of birthweight and subsequent weight gain, Day-2 body weight is predictive of weight (relative to agemates) through at least Day 30 postpartum for *A. cahirinus* pups reared with their biological mother. Thus, pups whose body weights were 1 SD or more below the mean on Day 2 also tended to weigh less than the group mean on Day 30. Early postnatal weight gain is not solely a function of birthweight alone, however. Rather, regardless of whether their Day-2 body weights are within or below the normal range, the rate at which pups gain weight is dependent at least partially on the characteristics of the lactating female with which they are raised. LBW pups reared by females who gave birth to N litters gain weight more rapidly than do their LBW littermates who remain with their biological mother. Alternatively, rate of weight gain by N pups fostered onto mothers of LBW pups is less than that of N pups raised by their own (N) mother. Thus females who produce LBW pups also support subnormal postnatal growth, whereas N mothers give birth to pups within normal weight ranges and support more normal weight gain postnatally. At present, the maternal mechanisms mediating production of LBW pups are not understood but are likely to differ somewhat from those involved in postnatal growth (postnatal maternal influences on pup weight gain are discussed further below).

Despite the influence of maternal characteristics on the rate of weight gain by LBW and N pups, in each instance, the mean weight of the fostered pups remained significantly different from that of their foster littermates. That is, LBW pups fostered onto N females did not catch up to the weight of the N pups who remained with those same N females. In a similar manner, N pups fostered onto LBW mothers remained heavier than their foster LBW littermates through 30 days of age. It therefore appears that body weight over the first 30 days postpartum is dependent upon an interaction between birthweight and postnatal maternal influences. Pups who are most likely to remain below the normal weight range are those from low birthweight litters who continue to be raised by their

TABLE 2. Summary of Results of Choice Tests for Bedding Soiled by Lactating versus Nonlactating Adult Females for both Low-birthweight (LBW) and Normal-weight (N) Pups Tested on Day 9 Postpartum.

	Pup Condition	
	LBW	N
$\bar{X}$ responses to bedding soiled by a lactating female	2.63	3.81
$\bar{X}$ responses to bedding soiled by a nonlactating female	2.06	2.25
$\bar{X}$ preference score	0.57	1.56
p (Wilcoxon's test)	n.s.	<.02
	(t = 42.5; n-ties = 15)	(t = 13.5; n-ties = 14)

own mother. Rate of physical development of such LBW pups is not immutable, however, since they display increased weight gain if provided with a more adequate maternal environment.

Specific maternal attributes that influence the rate of postnatal weight gain in *A. cahirinus* pups have yet to be elucidated. Nonetheless, one likely possibility is that milk provided by mothers of LBW pups is suboptimal, either in quality or quantity (or a combination of these two factors). Early maternal behavior unrelated to feeding per se, such as grooming and huddling with the neonates, could also have an effect on pup development (e.g., Smart, 1983; Smotherman & Bell, 1980). In this regard, it would be interesting to ascertain whether mothers of LBW pups interact differently with neonates (including N foster pups) than do mothers of N pups.

On both Day 2 and Day 9, low-birthweight (LBW) pups responded less discriminatively to conspecific chemical cues than did N pups. Although LBW pups chose home-cage bedding significantly more often than clean bedding on Day 2, their frequency of responses to the clean bedding material was nonetheless significantly greater than that observed for N pups. This increased frequency of clean-bedding choices by LBW pups occurred despite the fact that pups in this condition generally were *less* active than N pups (as indicated by their low frequency of total choice responses). During the Day-9 tests, LBW pups, in contrast with N offspring, displayed no statistically reliable evidence of being attracted to maternal pheromone.

Attraction to the odor of the home area, or to maternal pheromone, presumably functions to keep neonates within the vicinity of the mother and thereby allows for ready access to maternal resources necessary for continued survival (Porter, 1983). Any reduction in responsiveness to maternal pheromone during the preweaning period, such as displayed by the LBW pups in the present instance, would increase the likelihood of the pup becoming separated from its mother and thereby place the neonate in considerable jeopardy. It is also likely that the behavioral or perceptual deficits in LBW pups could have been an additional factor contributing to their reduced rate of physical growth. For example, diminished olfactory competence might adversely effect nipple localization or attachment, resulting in restricted milk ingestion. Therefore, consistent with the literature cited above regarding mortality rates among underdeveloped mammalian neonates and avian hatchlings, one would expect low birthweight to be correlated with heightened mortality in *A. cahirinus* living in their natural habitat.

Finally, the data presented above illustrate that the subject species, *A. cahirinus,* is well-suited for investigations into the developmental sequelae (both behavioral and physical) of low birthweight. Spiny mice are readily maintained and bred in the laboratory, and there is a wide range of body weights of the precocial neonates. LBW pups raised by their biological mother remain relatively small throughout the suckling period and evince deficiencies in early adaptive behavior. Rate of weight gain in LBW and N pups is susceptible to postnatal maternal influences, however, and it would be interesting to determine whether behavior development is likewise sensitive to similar environmental manipulations.

Notes

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